

its confidence explicitly, which constitute a different line of approach. Exploring such approaches may offer complementary insights into how models internally represent and express uncertainty.

Applicability to Black-box LLMs or Large Reasoning Models Our study is limited to open-source LLMs. Conducting mechanistic analyses on commercial black-box LLMs is not permitted due to access restrictions. Future work could explore alternative evaluation protocols or collaboration frameworks that enable partial interpretability analyses on such systems. In addition, recent studies (Mei et al., 2025; Zhang et al., 2025) have begun examining the internal states of large reasoning models for hallucination detection, suggesting a promising direction for extending our methodology to models with multi-step reasoning capabilities.

Ethical Considerations

This work analyzes the internal mechanisms of large language models using data constructed from Wikidata (Vrandečić and Krötzsch, 2014), which is released under the Creative Commons CC0 1.0 Universal license, allowing unrestricted use and redistribution of its data. All data are derived from publicly available resources, and no private or sensitive information about individuals is included. We used LLM tools for language polishing.

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